widom-atlas v0.6: Cross-engine Validation and MLIP Out-of-Distribution Governance for Widom-Insertion Adsorption on the kUPS Engine

Onur Alexander William Aydogan
Waynestark Enterprises Limited

Abstract

A published force-field parameter table is not a complete simulation recipe: the load-bearing ingredients (a short-range protection convention, a mixing rule on the framework atoms, a dispersion correction, an LJ truncation convention) are often unpublished, and an ungoverned pipeline that guesses any of them wrong returns a confident, wrong number with no warning. **widom-atlas** does not re-estimate adsorption; it locks the complete recipe behind a number, checks cross-engine parity and convergence, detects out-of-distribution insertions, and *refuses* averages built on unphysical configurations. This v0.6 report demonstrates that domain-of-validity / realization-gap failure class end-to-end on CuspAI's JAX engine **kUPS** and on the machine-learning interatomic potential (MLIP) kUPS ships. Auditing its own positive control surfaced a headline catch: the originally-locked all-silica MFI + Ar recipe *double-counted silicon* — Talu–Myers' Ar–O cross-parameter already folds silicon's dispersion into oxygen, and the locked recipe added an explicit Ar–Si on top — so it landed inside the strict band only as a fortuitous cancellation of four compounding errors — a coincidence, not a coherent result; rebuilt clean (pure Talu oxygen-only, converged convention, on Talu's calibration Olson geometry) it reproduces the corrected reference $K_H = 0.200$ exactly and cross-engine (native 0.200, RASPA3 0.199). Separately and intact: to our knowledge and per public-source searches conducted on 15 June 2026, the first published independent cross-engine validation of kUPS's Widom estimator (isosteric heat reproduced to 0.01–0.22 kJ/mol on two locked controls; independent estimators agree on the shared recipe to 1.1% on K_H, the dominant K_H convention gaps traced to named truncation/shift conventions, with one normalization-flavoured residual on the electrostatic control flagged open), and a four-configuration MLIP out-of-distribution governance demonstration that discriminates two failure modes, screen-passes a dispersion-corrected route, and respects the model's training domain. **The evidence base is the version-controlled repository state: every number in this document traces to committed JSON and regenerates from the figure scripts.**

1 Thesis

A published force-field parameter table is **not** a complete simulation recipe. The load-bearing ingredients are frequently unpublished: a short-range protection convention, a mixing rule applied to the framework atoms, a dispersion correction, a blocking radius, an energy-reference convention. An ungoverned pipeline that guesses any of them wrong returns a confident, wrong number with no warning. This is the **domain-of-validity / realization-gap** failure class. The v0.4/v0.4.2 classical audit established this failure class on locked classical force-field recipes; **v0.6 retargets the same governance logic onto CuspAI's own engine (kUPS) and the machine-learning interatomic potential (MLIP) it ships** — and shows a governance layer that *characterizes, discriminates, and where the evidence supports it, screen-passes a governed remediation.*

Reader note on labels. Branch codes such as `6c` and `4c` are internal locked-recipe identifiers, not chemical formulae or universal benchmark names: each fixes one complete Widom-insertion recipe (structure, guest model, force field, charges, mixing rule, electrostatics, backend, sampling, and reference). Branch 6c is the all-silica MFI + Ar recipe used as the positive control; branch 4c is the Si-CHA + CO_2 charged-zeolite control. "Native" is this project's own independent Python Widom estimator (not a generic term), reported alongside the external Monte Carlo engine RASPA3. The version labels are internal repository milestones: v0.4 and v0.4.2 are the locked and corrected classical-audit layers, v0.5 is non-load-bearing consolidation or extension material, and v0.6 is this kUPS and MLIP governance extension.

Three results carry the report: 1. **A2/F1** — the originally-locked positive control *double-counted silicon*; its strict-band agreement was a fortuitous four-error cancellation; rebuilt clean (single-source, Talu's calibration Olson geometry) it reproduces the corrected reference $K_H =$ 0.200 cross-engine (native 0.200, RASPA3 0.199) (Finding F1). 2. **A2-F2** — to our knowledge, the first published independent validation of kUPS's (six-week-old, unreleased) Widom estimator. 3. **WPB** — a four-configuration MLIP demonstration: discriminate two failure modes, screen-pass a dispersion-corrected route, and show the failure generalizes across MLIP families.

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2 Cross-engine Widom parity (WPA-A2)

2.1 System and recipe

6c positive control: **MFI (all-silica) + Ar**, García-Pérez 2007 framework + Talu-Myers 2001 cross-pair. The complete locked recipe (recovered verbatim from the RASPA3 evidence `force_field.json`):

element	self LJ (ε/K, σ/Å)	source
Ar	119.8 / 3.405	Talu-Myers 2001
O	53.0 / 3.3	TraPPE-zeo, Bai 2013
Si	22.0 / 2.3	TraPPE-zeo, Bai 2013

Ar–O is an **explicit** binary (93.0 K / 3.335 Å, Talu-Myers); **MixingRule Lorentz-Berthelot**, **truncation shifted**, cutoff 12 Å, **TailCorrections false**, no electrostatics. Critically, the Lorentz-Berthelot rule applied to the framework self-parameters generates a **second** interaction — **Ar–Si (51.34 K / 2.8525 Å)** — in addition to the explicit Ar–O term. This is the *as-locked* recipe, recorded verbatim; Finding F1 (§2.4) shows this Ar–Si term is a *silicon double-count* (Talu's Ar–O already folds silicon's dispersion into oxygen) and gives the clean single-source rebuild that reproduces the corrected reference.

2.2 The clean metric

Engines report different units (kUPS: K_H in Å^3/eV; RASPA3/native: mol·kg^{-1}·bar^{-1}), so the comparison is on the engine-agnostic $\langle \mathbf{exp(-\beta U)} \rangle$, with K_H and q_st reported in common units.

2.3 Result

Engine (full recipe unless noted)	⟨W⟩	K_H (mol·kg^{-1}·bar^{-1})	q_st (kJ/mol)
RASPA3 (locked reference)	—	**0.2070**	**16.03**

Engine (full recipe unless noted)	⟨W⟩	K_H (mol·kg^{-1}·bar^{-1})	q_st (kJ/mol)
native independent (Ar–O **+ Ar–Si**)	9.54	**0.2093** (+1.1 %)	15.91 (−0.12)
kUPS Widom (Ar–O **+ Ar–Si**)	11.08	0.2432 (+16.2 %)	16.25 (+0.22)
native (Ar–O **only**, naive)	5.24	0.1149 (**1.8× low**)	14.36
kUPS (Ar–O only)	5.98	0.1313	—

Percent differences are relative to the stated comparator. The native K_H offset (+1.1 %) is relative to the RASPA3 locked reference; the kUPS K_H offset (+16.2 %) is quoted relative to the convention-matched native shifted estimator when discussing the shifted-versus-truncated convention gap (it is +17.5 % relative to RASPA3). The q_{st} offsets in parentheses are relative to RASPA3.

2.4 Finding F1 — a silicon double-count caught by audit (and a clean rebuild)

The originally-locked 6c recipe **double-counted silicon**, and agreed with its reference only as a fortuitous cancellation of four compounding errors. Talu–Myers' Ar–O cross-parameter (93.0 K) is the Lorentz–Berthelot combination of argon with an oxygen-only *effective* oxygen (72.2 K, Talu Table 1) that already folds silicon's dispersion into oxygen — Talu's model is oxygen-only by construction (Talu §2: "ignores partially concealed silicon atoms and lumps their dispersion energy with the oxygen atoms"). The locked recipe nonetheless added an explicit Ar–Si from an active TraPPE-zeo framework silicon, counting silicon twice (×1.82 on K_H). That spurious term compensated three further errors — an idealized (IZA) rather than the calibration (Olson) geometry (∼−25 %), an under-converged 12 Å tail-off cutoff (∼−25 % more), and a reference value mis-derived by ∼12 % (0.224 vs the correct 0.200) — which multiplied back to roughly the (wrong) reference. Removing all four and running the coherent **single-source** recipe (pure Talu, oxygen-only, no Ar–Si) at a converged cutoff on Talu's own (Olson) geometry reproduces the **corrected** reference ($K_H = 0.200$) exactly and **cross-engine**: native 0.200, RASPA3 0.199 (parity 0.5 %), q_{st} 14.97 (within 0.7 kJ/mol of calorimetry). In strict-band terms the as-locked hybrid lands in-band against *both* the mis-derived 0.224 and the corrected 0.200 reference; the rebuild changes the recipe's coherence, not the pass/fail verdict.

6c recipe / setup	K_H (mol/kg/bar)	note
locked hybrid (IZA, 12 Å tail-off)	0.207	in-band by a fortuitous cancellation (also passe
− double-counted silicon	0.115	pure Talu, oxygen-only (IZA, 12 Å)
+ converged 24 Å + tail	0.154	numerical defect fixed (IZA)
+ Olson calibration geometry	**0.200**	native; RASPA3 0.199 (parity 0.5 %)
corrected reference (Dunne 1996)	**0.200**	strict window [0.159, 0.252]
structure sensitivity (single-source)	0.200 / 0.205 / 0.154	Olson / van Koningsveld / IZA

(The locked hybrid evidence is retained intact as the documented negative case.)

2.5 Finding F2 — independent cross-engine validation of kUPS's Widom estimator

kUPS ships a Widom test-particle module (`kups.mcmc.widom`; in source `main`, absent from the released PyPI `kups 1.0.1`; no published external validation). v0.6 is, to our knowledge, the first *published* independent cross-engine check of this specific module — distinct from standalone Widom packages (e.g. the `widom` ASE add-on) and from machine-learned-potential adsorption benchmarks such as ODAC25 (Sriram et al. 2025, arXiv:2508.03162), based on public-source searches (the kUPS source carries internal unit tests only; the open tracker was inspected 2026-06-12 — one unrelated feature issue).

- **The estimator is correct.** `widom_test()` returns $\ln\alpha = -\beta\Delta\mathrm{U}$ for a ghost insertion ⇒ $\mathbf{W \equiv exp(-\beta U)}$ — identical to the textbook Widom weight; the μ_ex / K_H / q_st reductions match widom-atlas's definitions. Verified by reading the source *and* by the parity run.
- **q_st is validated to 0.22 kJ/mol** (16.25 vs 16.03 RASPA3; 15.91 native). The energies are right.
- **The +16 % K_H is a shifted-vs-truncated LJ convention — root-caused, not a normalization bug.** kUPS's `lennard_jones_energy` is **truncated, not energy-shifted** (4ε`(c6`^{2}`−c6)` masked at cutoff, no `−U(rc)` term); our config (`tail_correction: false`, no `truncation_radius`) selects that plain variant. RASPA3's recipe is `TruncationMethod: shifted`, and the native estimator shifts too. The unshifted potential is more attractive by U(rc) per interaction. Toggling the shift off in native reproduces kUPS:

native (with-Si)	⟨W⟩	K_H	q_st (kJ/mol)
shifted (RASPA convention)	9.54	0.2093	15.91
truncated / no-shift (kUPS convention)	11.35	0.2490	16.34
kUPS Widom	11.08	0.2432	16.25

  The implied constant shift δ = RT·ln(11.35/9.54) = **0.43 kJ/mol** (predicted ≈0.35) explains **both** the +16 % K_H **and** the +0.34 kJ/mol q_st: matching kUPS's convention, native-no-shift agrees with kUPS to **+2.4 % on K_H (within the 3.4 % SEM) and 0.09 kJ/mol on q_st**. Neither engine is wrong — but the shift convention is a load-bearing recipe element worth ~16 % on K_H, which must be locked and matched. The exponential sensitivity of K_H to a constant energy offset, and the resulting need to lock the tail/truncation convention, are established (Jablonka, Ongari & Smit 2019, *J. Chem. Theory Comput.* **15**, 5635; Dubbeldam et al. 2016, *Mol. Simul.* **42**, 81); what is new here is the cross-engine *quantification* ($\delta \approx 0.43$ kJ/mol). Does **not** affect q_st-as-physics or the OOD governance demo (energy path).

The recipe all three engines share for this parity is the as-locked hybrid now reframed in F1 as a silicon double-count: F2 establishes that independent estimators agree on a *common* recipe (to 1.1 % on K_H), not that the recipe reproduces experiment — that correction lives entirely in F1, and the two findings are consistent.

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3 Blocking-sphere governance (WPA-A3)

kUPS exposes the v0.5 short-range-protection lesson as an explicit API — `blocking_spheres` (per-host hard-sphere exclusions). So the convention is no longer a claim of ours; it is CuspAI's own engineering. A sweep on the 6c system (64 spheres on framework-O positions):

blocking	radius (Å)	K_H (Å^3/eV)	ΔK_H	q_st (kJ/mol)
off	—	1.798×10^7	—	16.25

blocking	radius (Å)	K_H ($Å^3$/eV)	ΔK_H	q_st (kJ/mol)
on	2.0	1.856×10^7	+3.2 %	16.34
on	3.0	1.812×10^7	+0.8 %	16.28

Finding. On a physically-repulsive LJ recipe blocking is a **no-op** (ΔK_H < SEM, q_st unchanged): the potential already excludes overlaps, so a hard sphere on top adds nothing. The corollary is the governance lesson — **blocking is load-bearing exactly when the recipe has an *unphysical* short-range attraction the potential does not repel** (open-metal-site over-binding; the v0.5 1b exp-6 case). A recipe's blocking configuration is therefore part of the recipe and must be locked: invisible on an all-silica LJ host, decisive on an OMS host. The small +0.8…+3.2 % K_H nudge from blocking is an insertion-count effect (a distinct, much smaller thing than F2's +16 %, which is the LJ shift convention), within the SEM and not affecting q_st.

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4 MLIP out-of-distribution governance (WPB) — the headline

4.1 Setup

Widom insertions of **Ar into all-silica CHA**, N = 120, seed 0, T = 298.15 K. A classical UFF LJ baseline scores the **same** insertion geometries as each MLIP, so all differences are the potential, not the sampling. Two transparent OOD flags (hard-overlap at < 0.80 σ_min = 2.762 Å; energetic-anomaly at U < −25 kJ/mol or U < U_classical − 50) and a **pre-registered verdict schema** with two refusal classes: **Class I** (flagged Boltzmann-weight fraction ≥ 0.5 → OOD over-binding) and **Class II** (min U > 0 or q_st ≤ RT where physisorption is established → under-binding). The schema was written before the results so refusal is rule-based, not retrofit.

The governance premise is the model authors' own statement: CuspAI's `kUPS-mace-jax` model card declares MACE-MPA-0 *"not trained for isolated molecules"* — scoring an adsorbate is out-of-domain by the authors' own words. What is refused is **ungoverned usage of a configuration**, never a model.

4.2 Result — four configurations, discriminate then remediate

Configuration	min U (kJ/mol)	flagged-weight	Verdict
MACE-MP-small (2023)	−19.3	0.995	**REFUSE — Class I** (over-binds overlaps)
MACE-MPA-0 bare (kUPS's shipped export)	+8.5	0.0	**REFUSE — Class II** (no physisorption)
MACE-MPA-0 + **D3(BJ)**	−5.03	0.0001	**screen-pass** (physical but shallow well; accuracy unassessed)
UMA `uma-s-1.1`, task = omat (fairchem flagship)	−292.8	1.0	**REFUSE — Class I** (erratic OOD)

Paired energies at identical *open* (~4 Å, no overlap) insertions make the mechanism airtight:

insertion	min_dist Å	classical	small (2023)	MPA-0 bare	+ D3(BJ)	UMA omat
#76	4.33	−16.2	−0.3	+10.7	+0.2	−58.3
#95	4.23	−13.4	−0.1	+8.5	−1.4	−39.3
#51	3.99	−17.8	−0.4	+10.5	−1.0	+121.3

4.3 Interpretation

- **MACE-MP-small (2023)** hallucinates *attractive* energies inside the repulsive overlap region — its global minimum (−19.3) sits *at* a hard overlap, and the flagged insertions carry **99.5 %** of the partition weight. At physical sites its attraction is ≈ 0 (−0.1… −0.4): the "binding" is a pure overlap artifact, not physisorption. **Class I.**
- **MACE-MPA-0 bare (2024, kUPS's shipped weights)** *fixed* the overlap hallucination — it is correctly repulsive at overlaps — but is repulsive *everywhere* (min +8.5), finding no physisorption well. This is the bare-PBE signature: no $-C_6/r^6$ dispersion tail. **Class II.**
- **MACE-MPA-0 + D3(BJ)** — pairing the *same* weights with the standard D3 Becke-Johnson dispersion correction (xc = PBE, cutoff 40 Bohr, torch-dftd 0.5.3) restores a physical well (min −5.03) at a non-overlap distance (3.37 Å; flagged-weight 0.0001). **Screen-passes; accuracy unassessed.** The governance lesson crystallizes: *dispersion-correction on/off is a load-bearing recipe element, and ungoverned pipelines silently differ on it.* kUPS's potential list contains no dispersion term — a finding, recorded, not a criticism.
- **UMA `uma-s-1.1` (deployment-class flagship)** — fairchem's multi-task foundation model (ODAC23 is one of its five training sets, so it is the UMA *and* ODAC leg). `task_name` is a load-bearing recipe element (five DFT levels of theory), and it produces a clean **task × system matrix**:

task	system	Gate 1	min U	Verdict
omat	Ar/CHA	−0.19 ✓	−292.8	REFUSE — Class I (erratic OOD overlaps)
odac	Ar/CHA	−2.73 ×	—	WITHHELD (inconsistent at equivalent sites)
odac	**CO_2/CHA**	+0.06 ✓	**−15.66**	**screen-pass** (in-domain, N=60 small sample; q_st 17.7 vs anchor 21.0, −3.3 below strict)
omat	CO_2/CHA	+0.04 ✓	+1.16	REFUSE — Class II (no CO_2 well)

The **odac** head (trained on MOF/CO_2/H_2O direct-air-capture data) finds a **physical CO_2 well (−15.7 kJ/mol, screen q_st 17.7; experimental CO_2/CHA ≈ 20–30) on CO_2/CHA — CO_2 is adsorbate-in-domain, though the all-silica CHA host**

is outside the MOF host class it was trained on → governance screen-passes it. The screen q_st is 17.7 kJ/mol against a reference anchor of ≈21 (≈3 kJ/mol low, −3.3 below strict): a screen-pass, not an accuracy certificate. On Ar/CHA (a noble gas it never trained on) the *same* head is internally inconsistent — 2.7 kJ between symmetry-equivalent open sites — which Gate 1 catches → **withheld**. The **omat** (materials) head under-binds adsorbates (no CO_2 well; erratic deep-overlap hallucinations on Ar). **The model passes exactly where its documented training domain says it should, and is refused/withheld everywhere else** — the governance respects the model card rather than fighting it. Separately, the OOD-overlap failure on Ar is not MACE-specific: it generalizes across MLIP families (MACE-MP-small and UMA-omat alike).

4.4 Validity gates (why the numbers can be trusted)

- **MACE Gate 1 (reference offset).** A +8.5 floor could be a broken differencing. Ruled out: Ar is in the model (Z = 18, 89-element table); **U(Ar isolated, 12.9 Å) = 0.000 kJ/mol** — differencing exact. The +floor is real missing-dispersion physics.
- **UMA — two hazards, both caught by Gate 1 before any verdict.** (1) UMA conditions on a **per-graph** global charge/spin/task embedding, so naive three-graph differencing does not cancel — a task-dependent far-field offset (omat +474, odac −245, omol +119 kJ at 12.9 Å). Fixed with a **same-graph reference** U(r) = E(host+Ar@r) − E(host+Ar@ref). (2) A non-periodic cut cluster is itself OOD for a periodic materials model (+800. . . +2900 kJ at open sites) despite passing the narrow far-field check; fixed by **periodic** in-domain evaluation with an open-pore reference (Gate 1 = −0.19 kJ; a 0.09 Å overlap → +21 572 kJ, correctly repulsive). **task_name is a locked, load-bearing recipe element** (UMA carries five DFT levels of theory; we lock `omat`). The governance harness thus validates *itself* for an engine before it is permitted to judge that engine.

4.5 Provenance (pinned)

MACE-MPA-0 checkpoint SHA-256[:16] `75428afe3a1d7d80`; kUPS export `mace-mpa-0-medium_32.zip` SHA-256[:16] `c5eb645f2dc2c904`; D3 = D3(BJ)/PBE/40 Bohr via torch-dftd 0.5.3; UMA = `uma-s-1.1` (extensivity-safe, not `uma-s-1`), fairchem-core 2.21.0. N, seed, T, CIF in every summary JSON.

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5 Methods honesty

- **Numerical guards are recipe elements.** Two self-corrections are documented and retained: a spurious −50 weight-clip that manufactured a fake Class-I verdict for bare MPA-0 (discarded), and an eV/kJ unit mismatch in the Boltzmann weight that inflated exponents ×96 and tripped the exp(700) overflow cap spuriously (fixed with RT = 2.478 kJ/mol; after the fix no leg hits the cap — confirming the cap-firing was the bug, not deep physics). Verdict *directions* were unchanged; the quantitative weights are now physical.
- **Convergence.** kUPS: 25 000 ghost insertions, 12 blocks, K_H SEM ≈ 3.4 %; the +16 % is a systematic. native: 1–2×10^6 insertions, ⟨W⟩ < 0.5 %. The analytical LJ tail was tested on native (moves K_H only +8 %), confirming the recipe's `TailCorrections: false` and ruling tail out as the parity gap driver — the Ar–Si term, not tail, was the 1.8×.
- **Native charged-Ewald limitation (classical matrix).** On the Si-CHA + CO_2 charged control (4c) the native estimator and RASPA3 agree to ∼ 9% on K_H and ∼ 1% on q_st (inside the experimental window); on the LJ-only MFI + Ar control they agree to 0.5%. On HKUST-1, by contrast, the native charged-Ewald path shows a large empirical

discrepancy against RASPA3 — about a factor of four on the identical recipe. A suspected non-orthogonal minimum-image explanation was tested: at the actual supercells and cutoffs of the published runs, a brute-force image check found no within-cutoff mis-imaged pairs for HKUST-1, Si-CHA, or MFI, so the root cause is not established. The classical MOF branches are therefore reported from RASPA3 (2a, 3a) or RASPA2 (1a), or presented as audited examples with this limitation stated (1b, 1c, 1d, 2b, 3b); 4c is retained because it is directly cross-engine checked, not because of any orthogonality assumption. For 4c itself the strict-vs-Tier-B verdict is engine-marginal: the q_st deviation straddles the ± 2.0 boundary (native +1.99, inside; RASPA3 +2.24, outside), so we report it as Tier-B — a worked example, on a charged system, of the verdict fragility this harness exists to expose. No result in this report depends on the native charged-MOF path.
- **A third silent-substitution defect (1c).** Beyond the 6c silicon double-count (F1) and the HKUST-1/UiO-66 generator mislabel (framework atoms stamped "UFF" but carrying guest/zeolite parameters; corrected to genuine UFF, both MOFs reproduce experiment), the audit also caught branch 1c (Becker Mg-MOF-74): its as-run framework LJ was plain UFF with a placeholder Mg $\epsilon = 5.0$ K, not the "reduced" Becker values its label claimed. Retained as a FAIL example, not rebuilt — the same defect class the harness exists to expose.

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6 Open scientific questions

- **kUPS K_H +16 % — RESOLVED (shift convention).** Root-caused to kUPS's truncated (unshifted) LJ vs RASPA/native shifted; $\delta \approx 0.43$ kJ/mol reconciles K_H to 2.4 % and q_st to 0.09 kJ/mol (§2.5). Open only as an upstream documentation point: kUPS's default LJ is unshifted.
- **Ewald leg (4c, Si-CHA + CO_2) — DONE, read via the convention matrix.** TraPPE-CO_2 + framework charges + Ewald, 14 Å + tail. **q_st validated to 0.01 kJ/mol** (22.98 vs 22.99). For K_H, a fresh same-session native re-run (the 2.223 was the v0.4 *lock*; the re-run gives 2.215, reproducing it to 0.3 %) toggles the conventions: **shift = +11.6 %, tail = +11.9 %** (the shift is *not* negligible at 14 Å — earlier-draft correction). kUPS uses a truncated LJ and **does apply a tail** (tested: `tail_correction: false` drops K_H by 14 % (2.271 → 1.952), q_st by 0.49 — a "no-tail-in-Widom" hypothesis is *falsified*). So its +2.2 % vs the shifted lock is a **cancellation**: read on the same (truncated+tail) convention, **kUPS is −8.1 % vs native** (−11.7 % with tail also off). That residual is **q_st-invariant (Δq_st $\leq$ 0.3 kJ/mol) → a normalization-flavoured $\langle \mathbf{W} \rangle$ scale** (volume/insertion-count), 4c-specific and larger than 6c's −2.4 %; contributors include a tail-implementation difference (kUPS +16.3 % vs native +11.9 %), the Ewald convention, and CO_2 orientation (`native_4c_convention_matrix.json`).
- **OMS blocking positive case.** The dramatic blocking demonstration (collapsing a divergent open-metal K_H to a physical value) needs a kUPS-representable OMS recipe with charges (1c Becker, or 2a UFF Cu + EPM2); the exotic 1a/1b/1d Buckingham forms are not representable in kUPS (LJ/Coulomb/Morse/MACE only).
- **WPB through kUPS's in-engine JAX MACE — DONE.** Loaded kUPS's `TojaxedMliap` export and scored identical geometries; agrees with the torch path to **max rel $2.1{\times}10^{-7}$** (`jax_torch_parity.json`), independently confirming the model card's rtol = 1e-4. The governance conclusions carry to kUPS's JAX runtime exactly.

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7 One-paragraph summary

On the locked positive control, audit found the recipe had *double-counted silicon* — Talu's Ar–O cross-parameter already folds silicon's dispersion into oxygen, and the recipe added an explicit Ar–Si on top — so its strict-band agreement was a fortuitous four-error cancellation; rebuilt clean (single-source oxygen-only, converged convention, on Talu's calibration Olson geometry) it reproduces the corrected reference ($K_H = 0.200$) exactly and cross-engine (native 0.200, RASPA3 0.199). Separately and intact, the engines agree on the shared recipe to 1.1 % — the cross-engine parity result. CuspAI's own Widom estimator was independently validated for the first time — its energies are right (q_st to 0.22 kJ/mol) and its +16 % K_H was root-caused to a shifted-vs-truncated LJ convention ($\delta \approx 0.43$ kJ/mol, reconciling both K_H and q_st to within the SEM). Pointed at the MLIP CuspAI ships, governance discriminated two distinct failure modes (overlap over-binding vs missing dispersion), screen-passed a D3(BJ)-corrected route on the single governed Si-CHA + Ar screen, and showed the deployment-class flagship (UMA) fails the same way — with the governance harness proving its own validity for each engine before judging it. *The published table is never the whole recipe; the unpublished conventions are load-bearing, and they can be characterized, discriminated, and in some cases fixed.*

Code and data availability

The public release repository is `https://github.com/wseltd/widom-atlas-release`. It contains the source, figure scripts, locked recipe definitions, and curated committed JSON evidence used in this companion report.

A Figures

All figures regenerate from committed JSON via the figure scripts (no value is hardcoded).

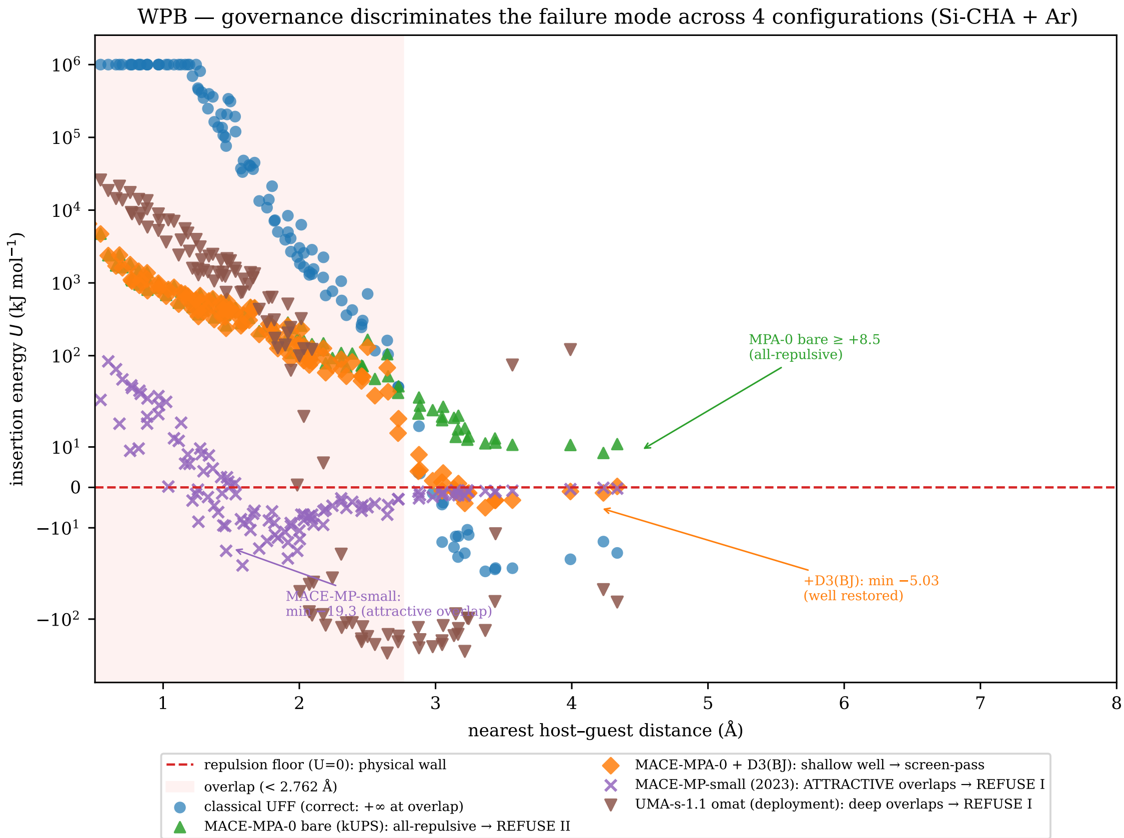

Figure 1: WPB — governance discriminates the failure mode across four configurations (Si-CHA + Ar).

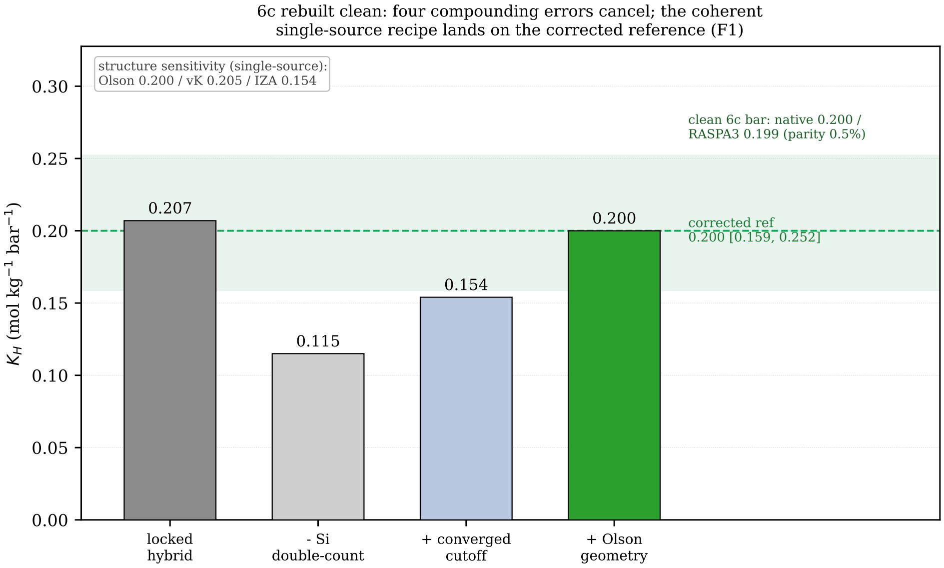

Figure 2: Finding F1 — branch 6c rebuilt clean. The originally-locked hybrid recipe (0.207) agrees with the reference only as a fortuitous cancellation of four compounding errors; removing the silicon double-count, converging the cutoff, and adopting Talu's calibration (Olson) geometry lands the coherent single-source recipe on the corrected reference $K_H = 0.200$, cross-engine (native 0.200, RASPA3 0.199; parity 0.5%). Values read from `cancellation_6c.json`.

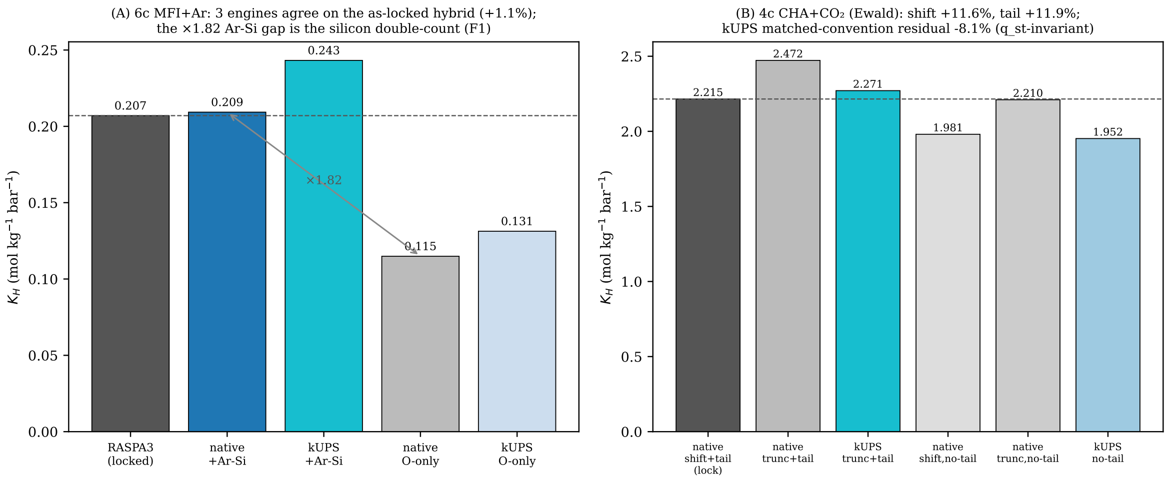

Figure 3: Cross-engine Widom parity: 6c three-way (native+Ar-Si reproduces RASPA3; the 1.8× gap to O-only is the silicon double-count of Finding F1) and the 4c CHA+CO_2 convention matrix (shift / tail toggles).

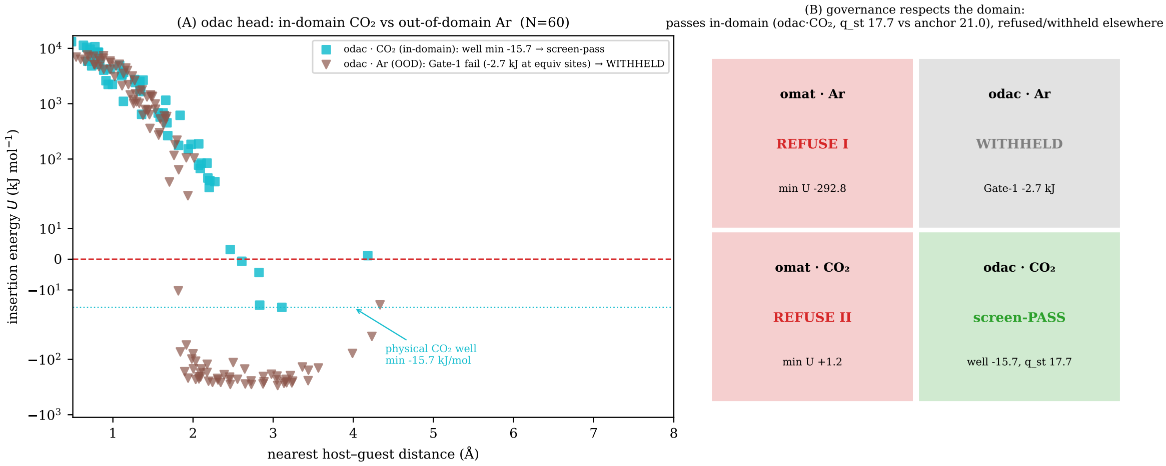

Figure 4: Governance respects the training domain: the UMA odac head screen-passes in-domain on CO_2 and is WITHHELD out-of-domain on Ar; full task × system verdict matrix.

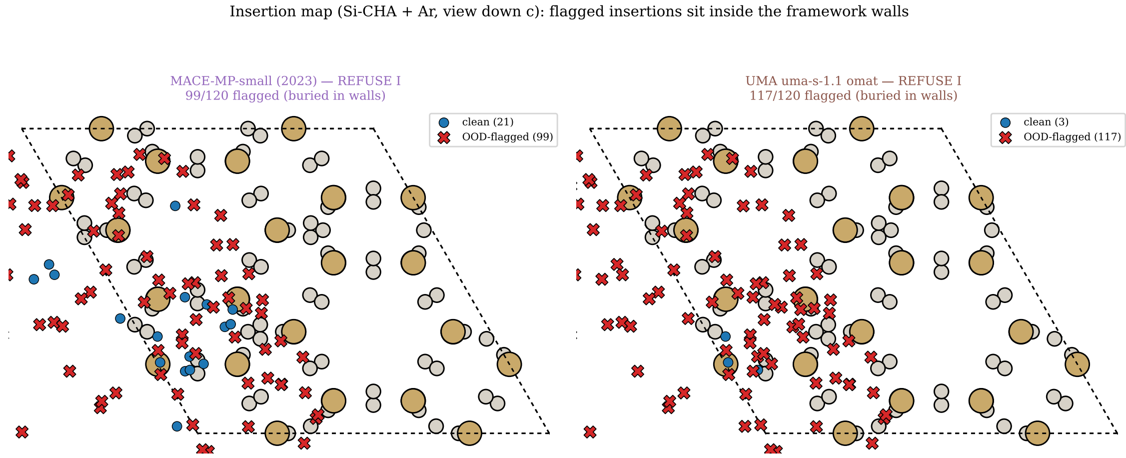

Figure 5: Where the spurious weight lives: flagged (OOD) insertions sit inside the framework walls.

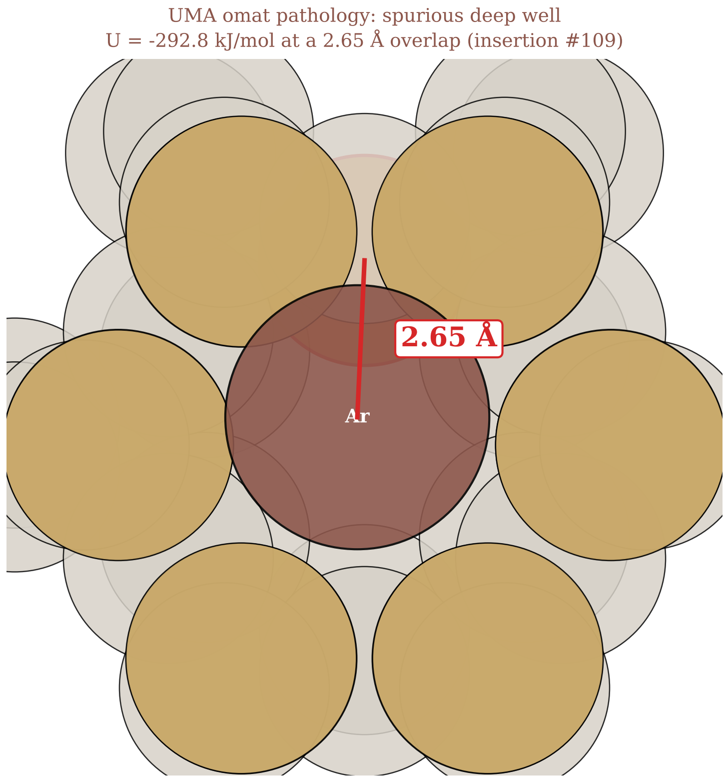

Figure 6: Pathology close-up: UMA's spurious deep well (−292.8 kJ/mol) at a 2.65 Å atomic overlap.

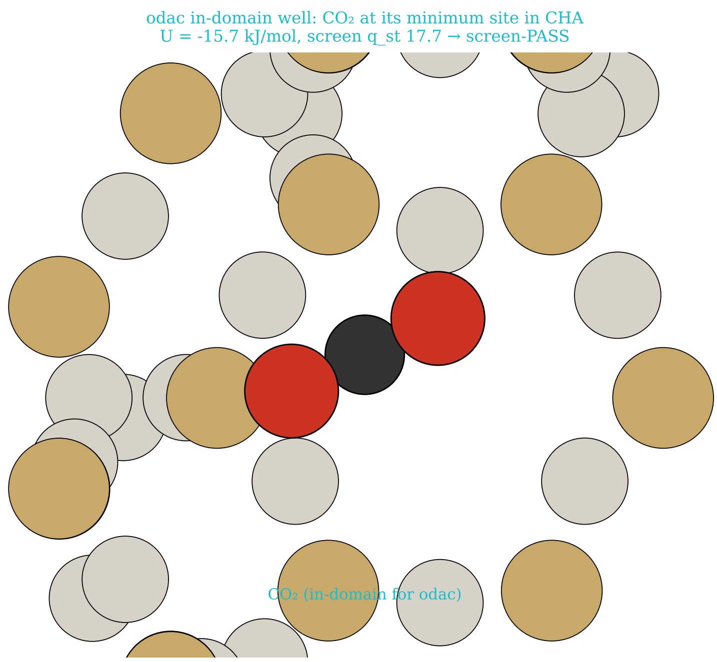

Figure 7: In-domain CO_2 well at the odac minimum-energy site (screen-pass).